

ME 524: Additive Manufacturing

Course Code	ME 524
Course Name	Additive Manufacturing
L-T-P-C	3-0-0-3, Elective
Prerequisites	None
Intended for	B.Tech./M.Tech/Ph.D
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

- **Introduction to Additive Manufacturing, Data formats and Preprocessing:** History, Comparison, Evolution, Methodology, Process chain, Classification, AM file formats, Part orientation, Support structure generation, slicing, Contour and tool path generation and build file preparation. [05 Hours]
- **Additive Manufacturing Methods:** AM equipment and Materials including Bio-active materials, Vat photo polymerization, Material jetting, Binder jetting, Material extrusion, Sheet lamination, Powder Bed fusion, Direct energy deposition, Thermal spray direct writing, Liquid phase direct deposition, PCB printing, Bioprinting, Concrete 3D printing, 4D printing and Hands on experience on the available AM machines. [22 Hours]
- **Design for AM:** DFMA, Part replacement, Adapt for AM, Design guidelines (part, support structure, Hole size, layer and wall thickness, residual stresses, optimization), Case studies. [05 Hours]
- **AM equipment and materials:** Laser, Electron beam, Arc, Beam, In situ monitoring, Polymer, Metal, Ceramics, Recent advances, Powder production and characterization. [05 Hours]
- **Post processing, Safety considerations, applications and Industry 4.0:** Quality evaluation, Surface finish and geometry improvement, Potential hazards, Powder and chemical hazards, Applications – Aerospace, Defence, Automobile, Biomedical, Rapid tooling, Reverse engineering, Industry 4.0 and future scope of AM. [05 Hours]

Text Books

1. Andreas Gebhardt, Jan-Steffen Hötter, **Additive Manufacturing: 3D Printing for Prototyping and Manufacturing**, Hanser Publications, 2016.
2. Chua Chee Kai, Leong Kah Fai, **3D Printing and Additive Manufacturing: Principles and Applications**, World Scientific, 2014.
3. Hod Lipson, Melba Kurman, **Fabricated: The New World of 3-D Printing**, Wiley 2013.
4. Patri K. Venuvinod, Weiyin Ma, **Rapid Prototyping - Laser-based and Other Technologies**, Kluwer Academic Publishers, 2003.